

2MVA Problem Sheet 1
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1. Calculate all second order partial derivatives of $z = f(g(x, y), h(x, y))$ with respect to x and y , given that $g(x, y) = 2x + y$ and $h(x, y) = 2x - y$, assuming that f is sufficiently smooth.

[20]

2. The function $z(x, y)$ is defined by the following equation

$$\cos(xy) + \ln(z) = yz - 1/2.$$

Find the partial derivatives z_x and z_y and calculate their values at the point $(\pi, 1/2, 1)$.

[15]

3. Consider $f(x, y, z) = e^{xyz}$ at point $\mathbf{a} = (1, 2, 3)$, find

(i) The direction and rate of steepest ascent for f at $\mathbf{a}$.

(ii) The direction and rate of steepest descent for f at $\mathbf{a}$.

(iii) Write the equation for a curved level surface passing through the point with the normal $\nabla f(\mathbf{a})$.

[20]

4. In the partial differential equation

$$(x + y) \frac{\partial z}{\partial x} - (x - y) \frac{\partial z}{\partial y} = 0$$

change the variables $(x, y) \rightarrow (u, v)$, where $u = \ln \sqrt{x^2 + y^2}$, $v = \tan^{-1}(y/x)$.

[20]

5. Find

$$\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial^2 u}{\partial x^2}, \frac{\partial^2 u}{\partial y^2}, \frac{\partial^2 u}{\partial x \partial y}$$

for the following function: $u = \sin(x \sin^{-1} y)$.

[25]